

IBM Data Analyst Capstone Project

Stack Overflow Developer Survey Analysis

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EXECUTIVE SUMMARY



Key Findings:

- Analyzed Stack Overflow Developer Survey data
- Performed data wrangling and exploratory data analysis
- Visualized technology trends and developer preferences
- Built an interactive dashboard using Google Looker Studio
- Identified trends in programming languages, databases, compensation, and job satisfaction



INTRODUCTION



- The Stack Overflow Developer Survey:
 - provides insights into developer technologies, tools, compensation, and preferences across the global developer community.
- This project analyzes survey responses to identify trends in programming languages, databases, platforms, and job satisfaction.

PROJECT OBJECTIVES

- Collect and explore developer survey data
- Clean and prepare data for analysis
- Perform exploratory data analysis (EDA)
- Create meaningful visualizations
- Build an interactive dashboard
- Present findings and insights



DATASET OVERVIEW

Dataset Source:

Stack Overflow Developer Survey

<https://stackoverflow.blog/2024/08/06/2024-developer-survey/>

Key Features:

- Age
- Country
- Employment
- Programming Languages
- Databases
- Compensation
- Job Satisfaction



METHODOLOGY



Workflow:

- Data Collection
- Data Wrangling
- Exploratory Data Analysis
- Data Visualization
- Dashboard Creation
- Presentation of Findings

Tools:

Python, Pandas, NumPy, SQL, Matplotlib, Seaborn, Google Looker Studio (DataStudio, now).

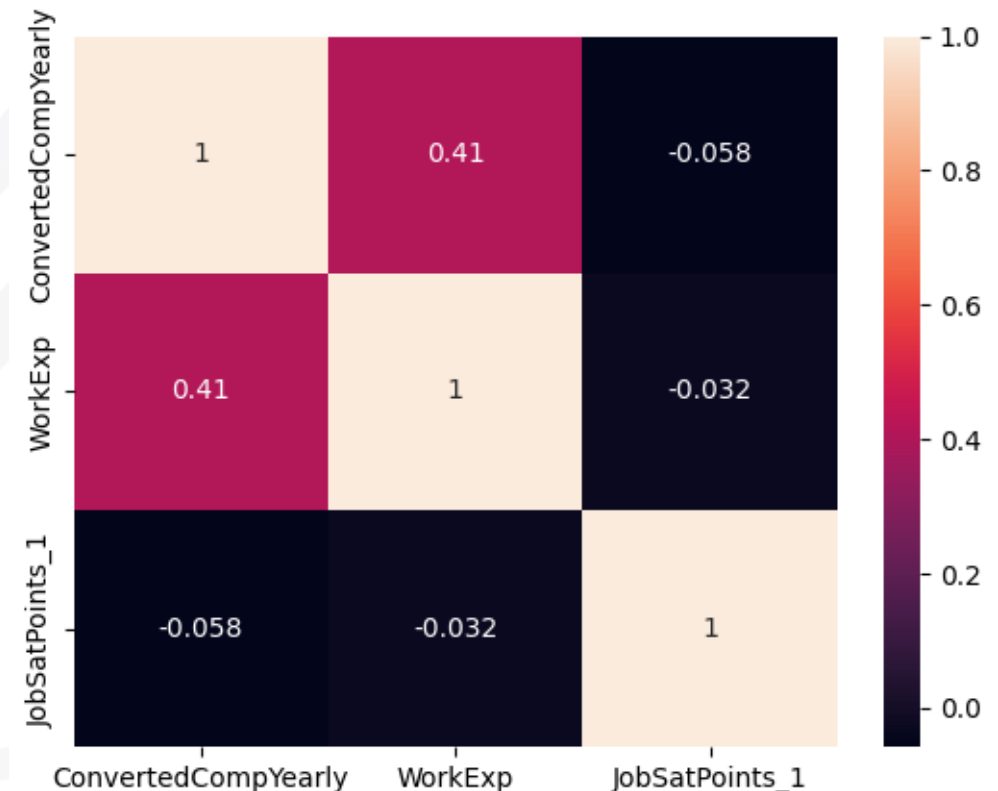


DATA WRANGLING

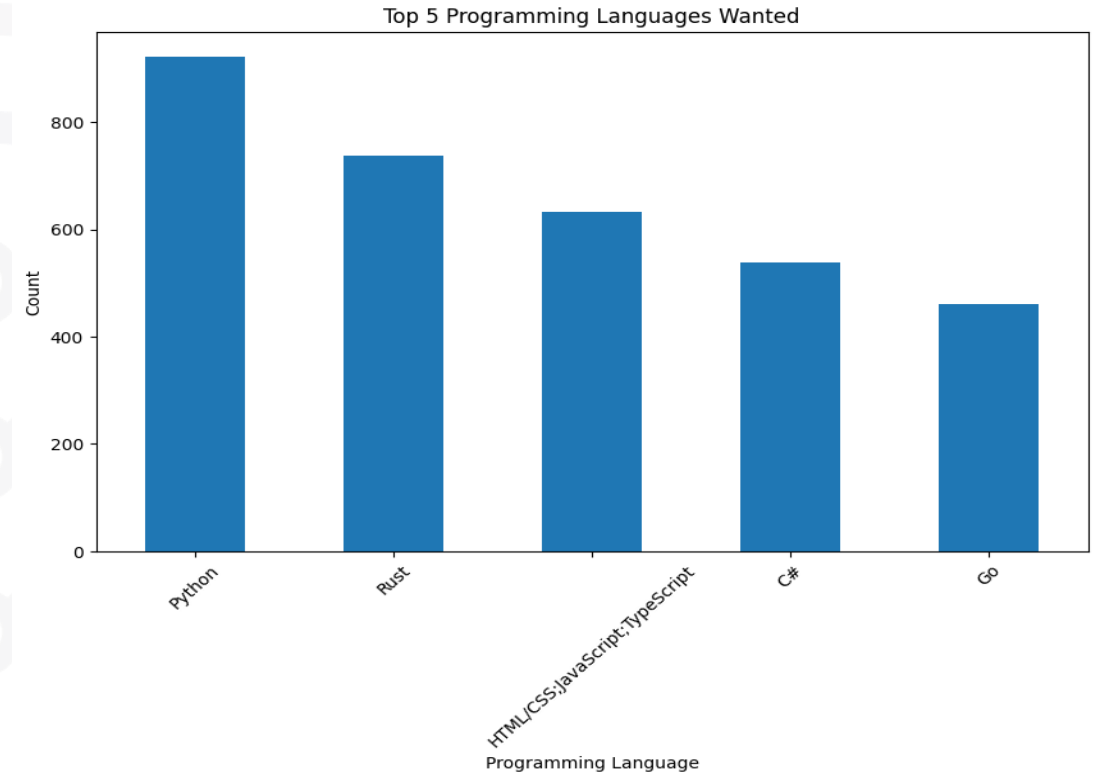
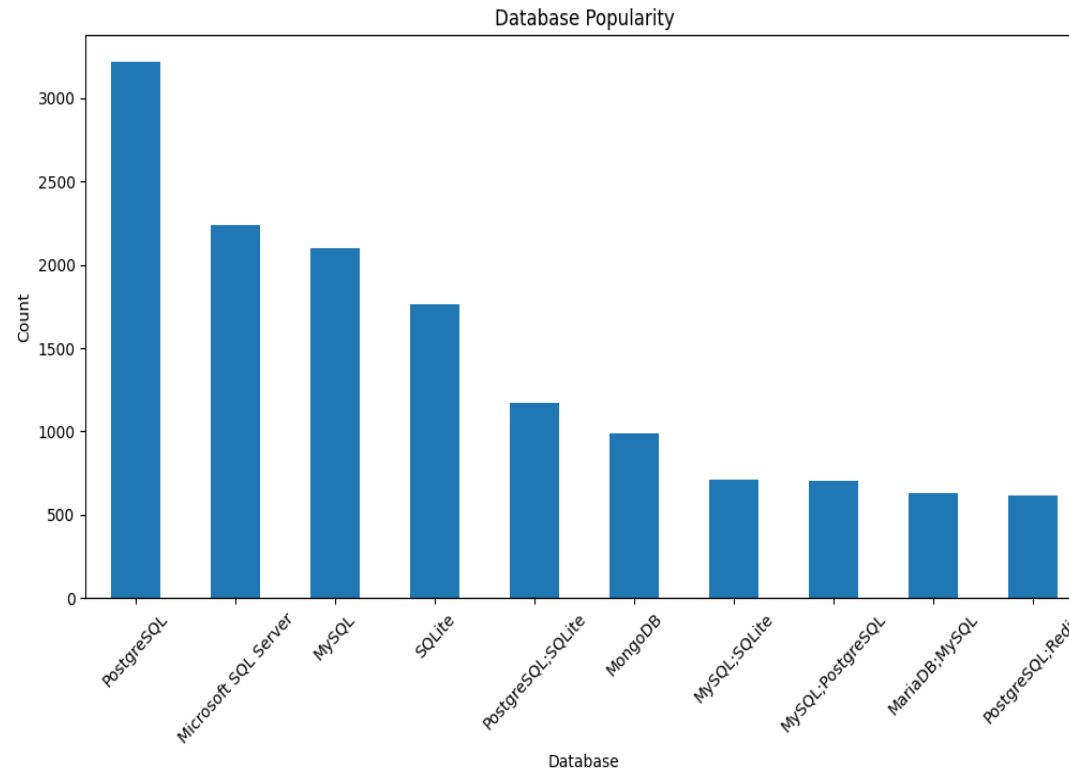
- Removed duplicate records
- Handled missing values
- Converted categorical age ranges into numerical
- Normalized compensation data
- Filtered invalid or extreme outliers

EXPLORATORY DATA ANALYSIS

- Compensation distribution showed significant variation
 - Compensation generally increased with experience
 - Job satisfaction varied across demographic groups
- Example chart for correlation heatmap:



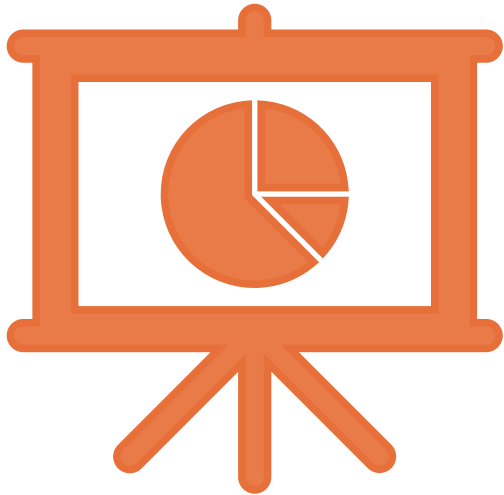
DATA VISUALIZATIONS



- Python, JavaScript, and SQL were among the most commonly used technologies.



Results



Visualization-Charts: Results in the following slides.

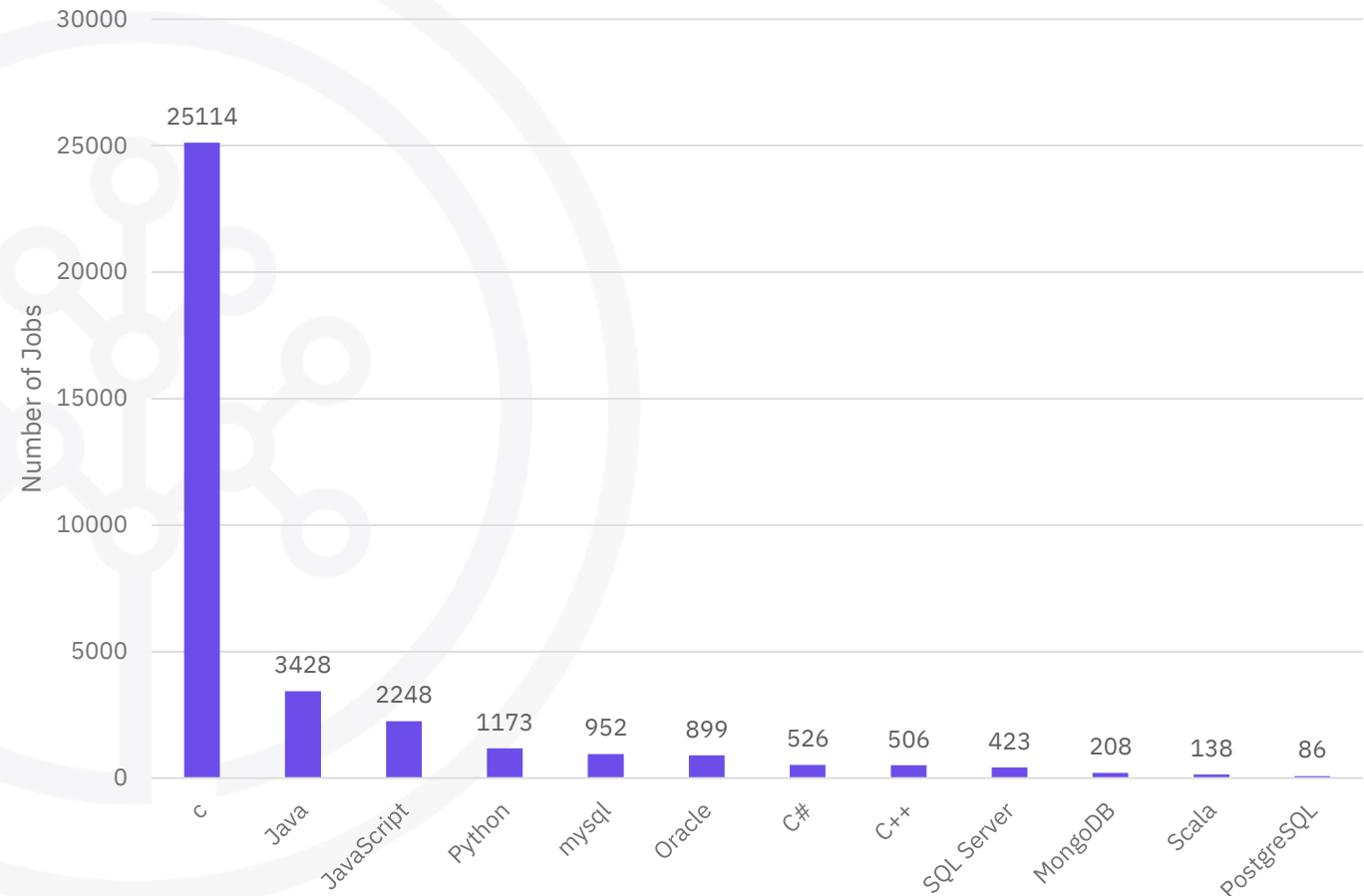


JOB POSTINGS

Job Postings Collected Using API.

- The analysis of job posting data shows that C had the highest number of job postings by a very large margin, with over 25,000 postings.
- Java followed with approximately 3,400 postings, while other technologies showed significantly lower demand.
- This indicates that low-level programming and system-oriented technologies continue to maintain strong industry relevance.

Top Technologies by Job Postings

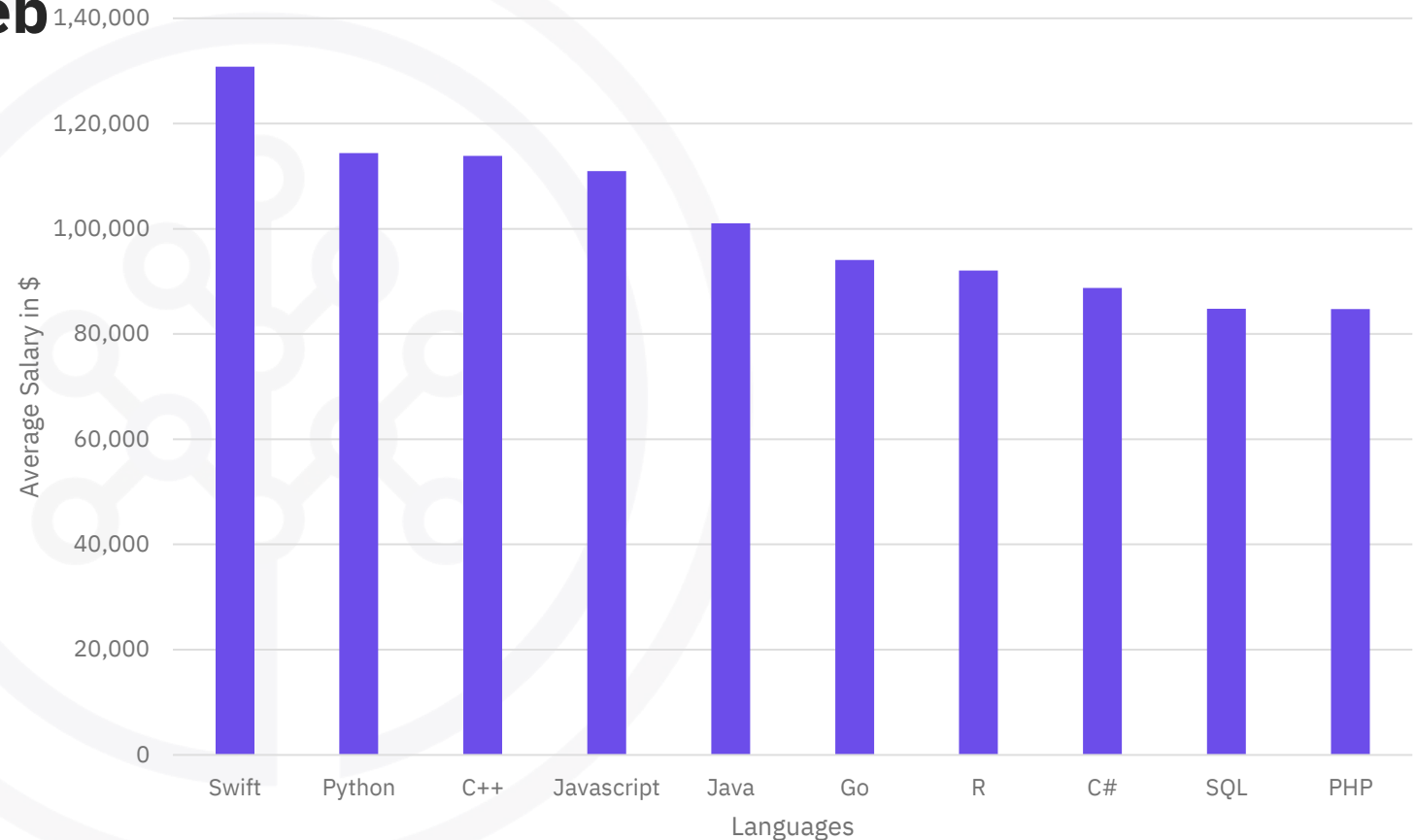


POPULAR LANGUAGES

Popular Languages Using Web Scraping.

- The salary analysis revealed that Swift had the highest average salary among the listed programming languages.
- Python and C++ followed closely with nearly equal salary levels, while JavaScript ranked third.
- These trends suggest that specialized and high-performance programming skills continue to offer strong compensation opportunities.

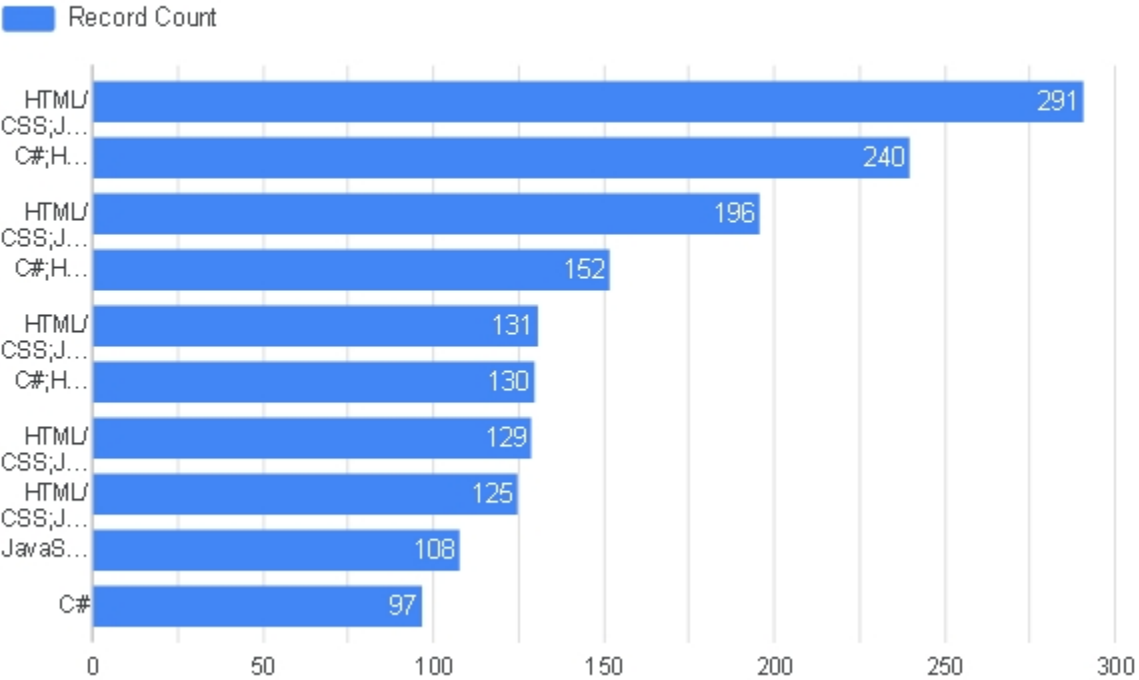
Popular Programming Languages by Salary



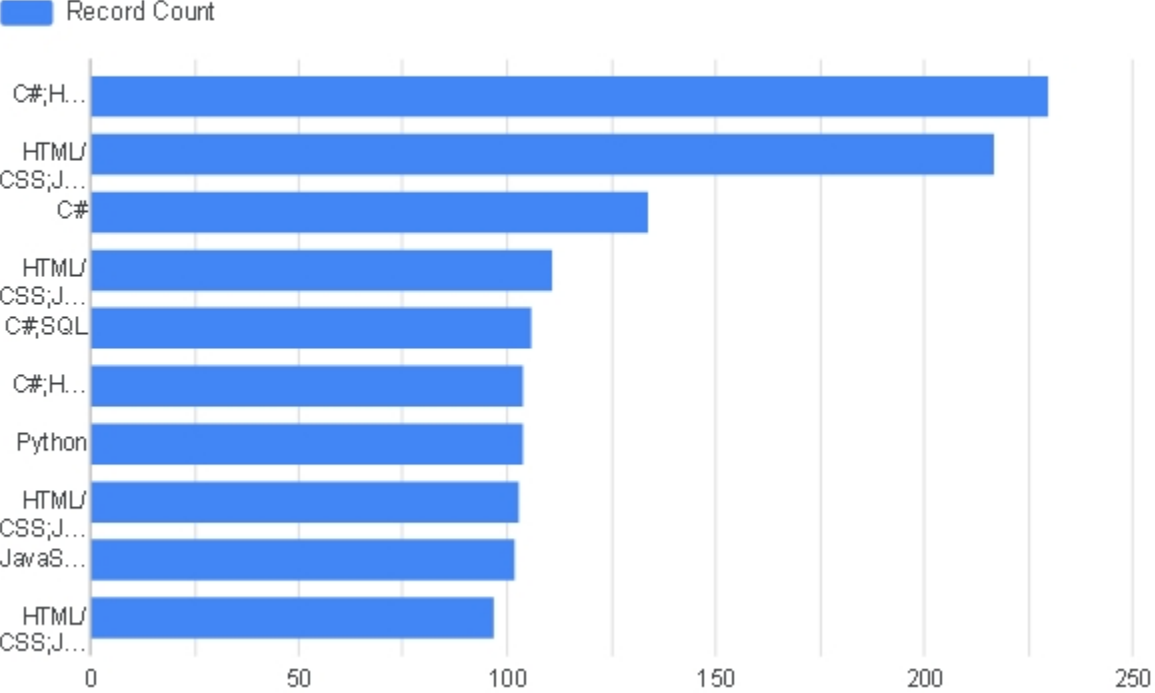
PROGRAMMING LANGUAGE TRENDS

Summary: Programming language trends were dominated by HTML/CSS, JavaScript, TypeScript, and C#, reflecting strong demand for modern web and full-stack development technologies.

Top 10 Languages Used



Top 10 Languages Desired Next Year



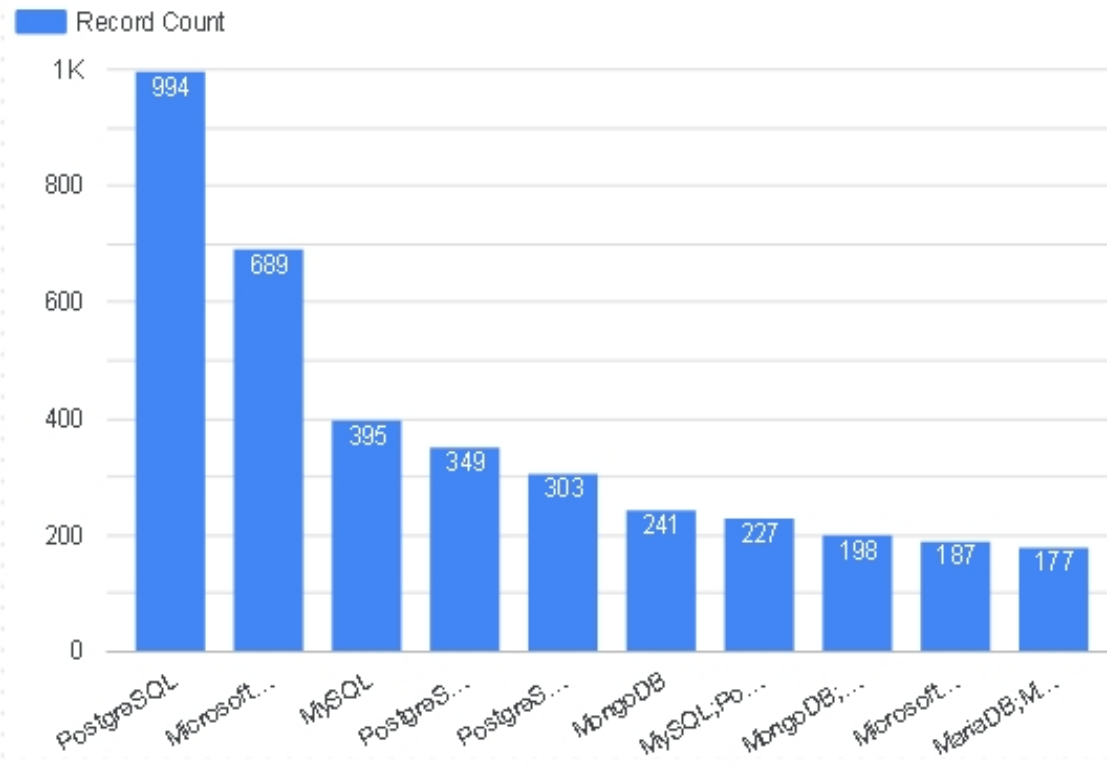
PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

- HTML/CSS, JavaScript, and TypeScript remain core technologies for modern web development.
- C# and SQL continue to show strong relevance in enterprise and backend systems.
- Developers are increasingly interested in full-stack development skill combinations.
- Future demand suggests continued growth in scalable web technologies and modern application frameworks.

DATABASE TRENDS

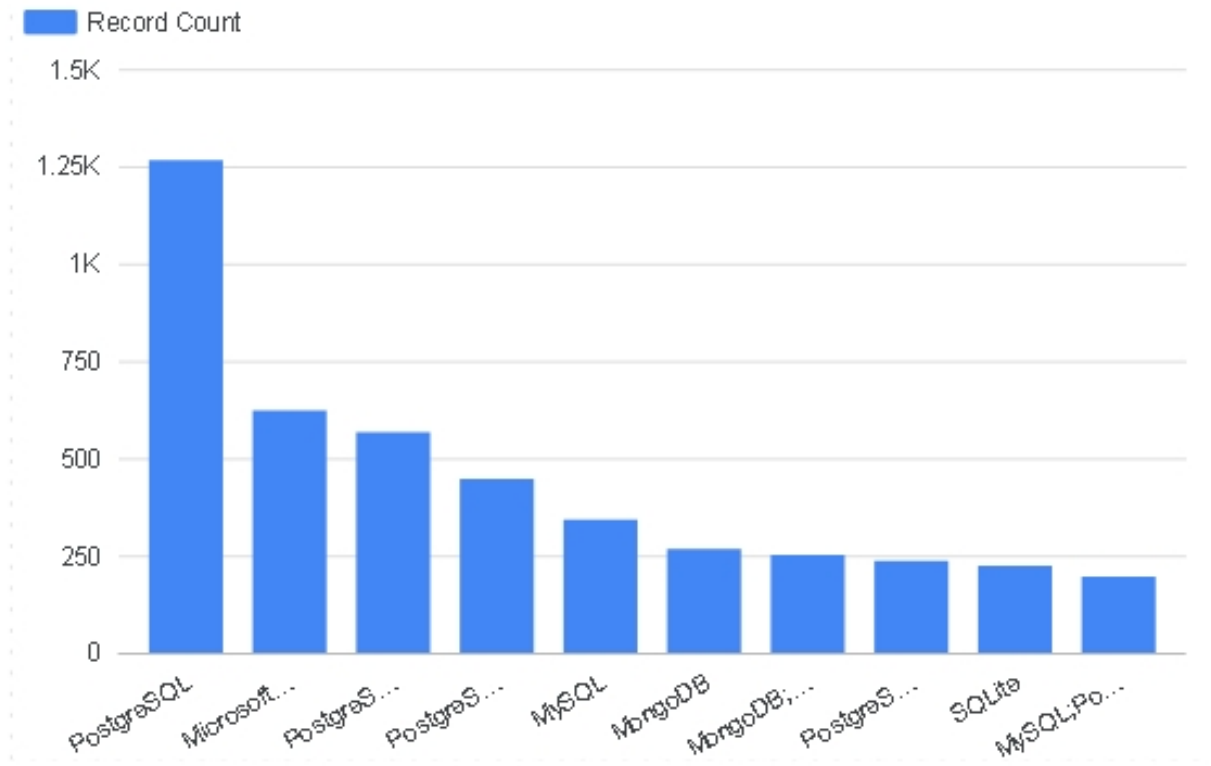
Database trends showed PostgreSQL, Microsoft SQL Server, and MySQL as the most widely used and desired database technologies among developers.

Top 10 Databases Used



Current Year

Top 10 Databases Desired Next Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

- PostgreSQL showed the highest popularity in both current usage and future preference trends.
- Microsoft SQL Server continues to maintain strong relevance in enterprise environments.
- MySQL remains widely used due to its reliability and accessibility.
- Future trends indicate growing interest in lightweight and scalable database combinations such as PostgreSQL and SQLite.
- Database management skills continue to be essential across software development and data analytics roles.



DASHBOARD



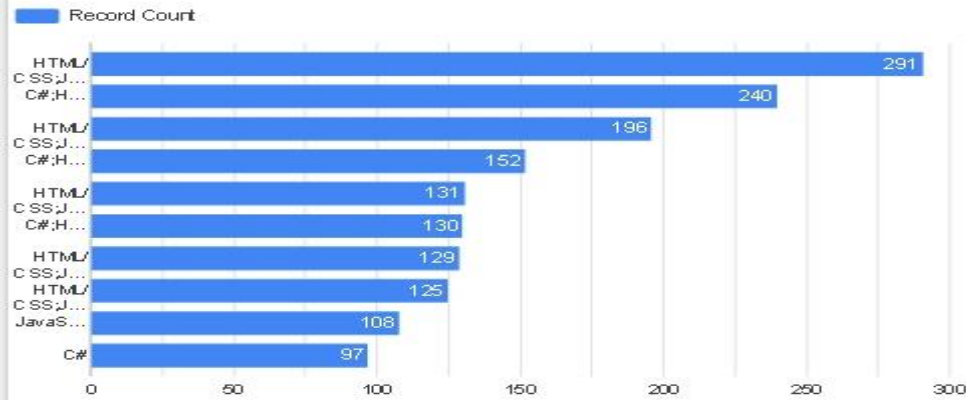
Dashboard results in the following slides.



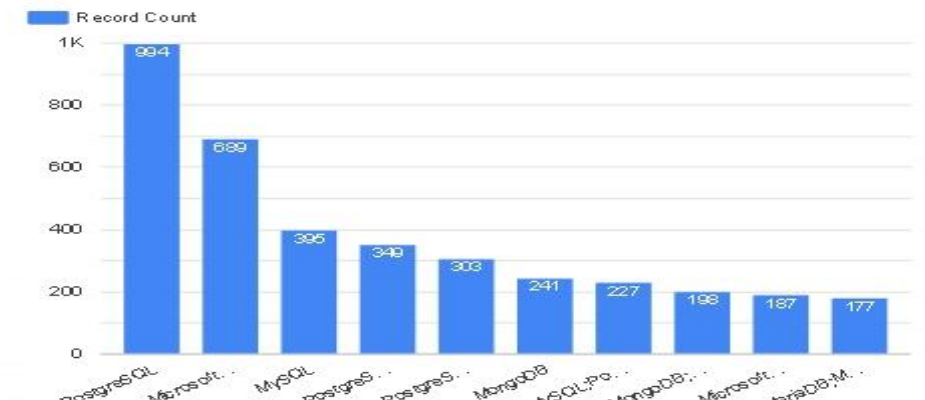
DASHBOARD TAB 1: Current Technology Usage

Summary: The dashboard highlighted the most commonly used programming languages, databases, platforms, and web frameworks among developers.

Top 10 Languages Used



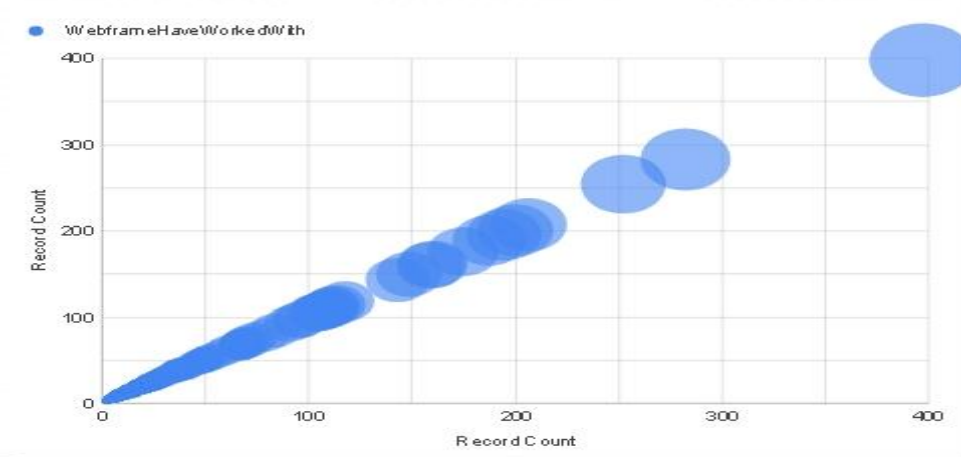
Top 10 Databases Used



Top 10 Platforms Used



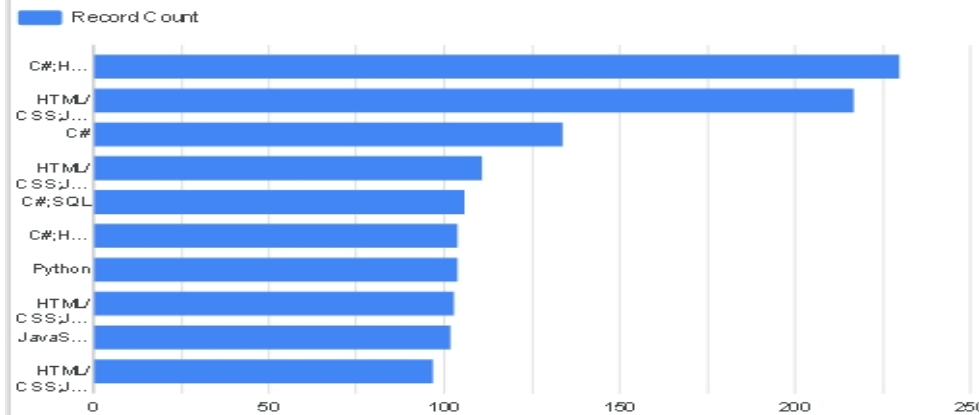
Top 10 Web Frameworks Used



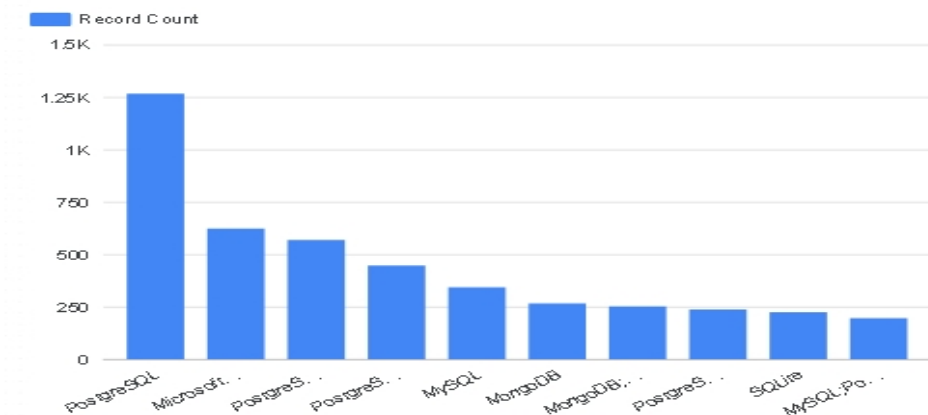
DASHBOARD TAB 2: Future Technology Trends

Summary: The dashboard revealed growing interest in modern full-stack technologies, scalable databases, and future-ready development platforms.

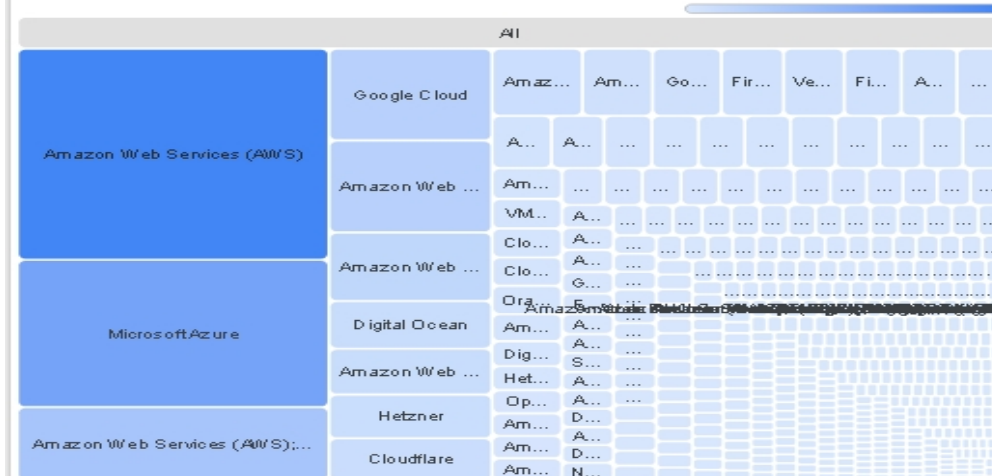
Top 10 Languages Desired Next Year



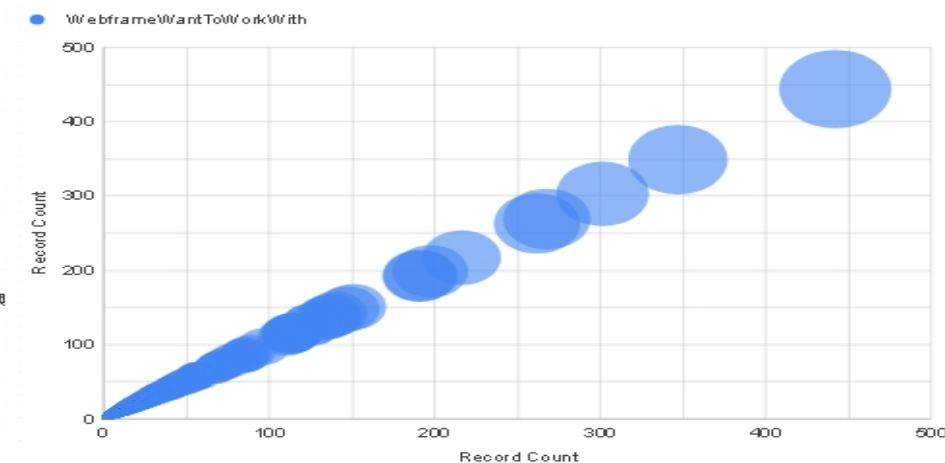
Top 10 Databases Desired Next Year



Top Desired Platforms



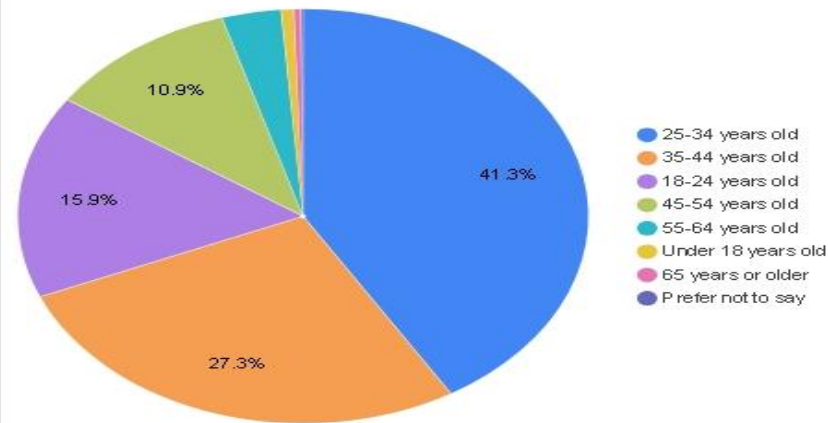
Top Desired Web Frameworks



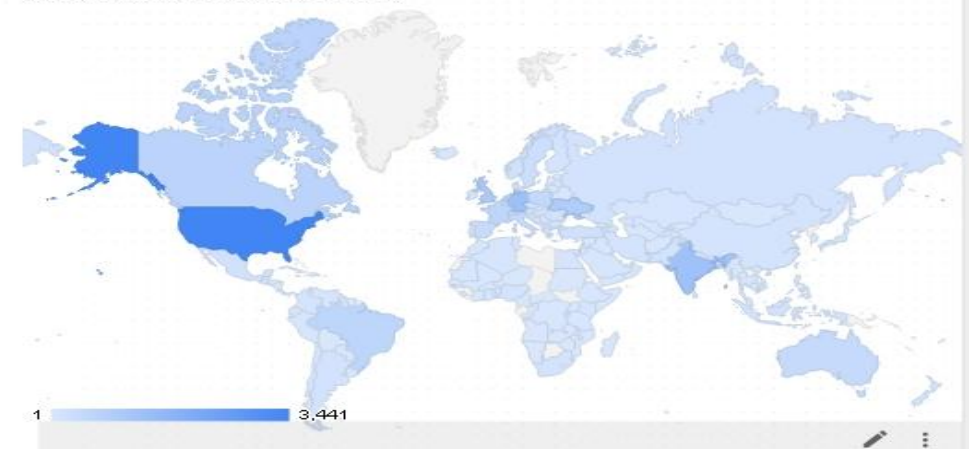
DASHBOARD TAB 3: Demographics

Summary: The dashboard provided insights into developer demographics, including respondent distribution by country, age, and education level.

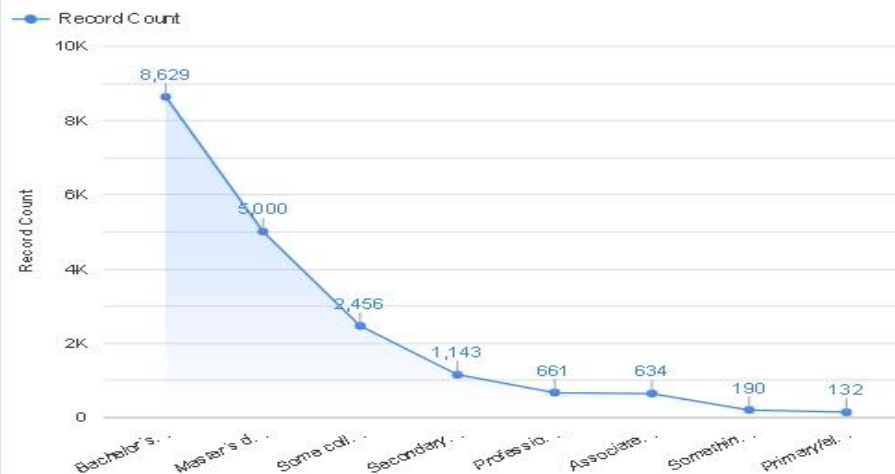
Respondents by Age



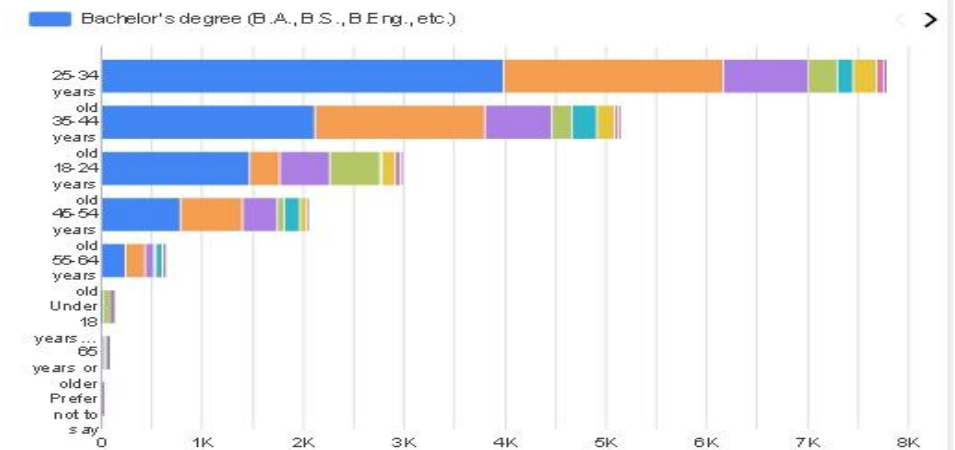
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age and Education Level



DISCUSSION



- The dashboard highlights the technologies most commonly used by developers, including programming languages, databases, platforms, and web frameworks. Web development technologies and relational databases showed particularly strong adoption trends.
- The dashboard indicates increasing interest in modern full-stack development technologies, scalable databases, and cloud-compatible platforms. Technologies involving TypeScript, C#, and PostgreSQL showed strong future demand among developers.
- The demographic dashboard provides insights into respondent distribution across countries, age groups, and education levels. The data reflects a globally diverse developer community with strong participation from professionally educated and early-to-mid career developers.



OVERALL FINDINGS & IMPLICATIONS

- Current technology usage trends showed strong dominance of web development technologies such as HTML/CSS, JavaScript, and TypeScript.
- Future trend analysis indicated growing interest in full-stack development technologies involving C#, SQL, and TypeScript.
- PostgreSQL emerged as the most widely used and desired database technology among respondents.
- The demographic analysis reflected a globally diverse developer community with varying educational and professional backgrounds.
- Dashboard insights suggest increasing industry demand for scalable, cloud-compatible, and modern application development technologies.
- Developers with skills in programming, databases, and full-stack technologies are likely to remain highly competitive in the evolving technology landscape.

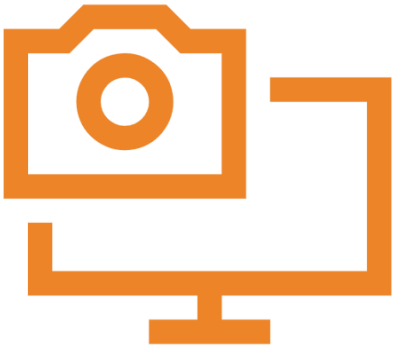
CONCLUSION



- This capstone project successfully demonstrated the complete data analysis workflow, including data collection, data wrangling, exploratory data analysis, visualization, dashboard creation, and presentation of findings.
- The analysis provided valuable insights into current technology usage, future technology trends, database preferences, developer demographics, and salary patterns using the Stack Overflow Developer Survey dataset.
- The project also highlighted the importance of data visualization and dashboarding techniques in transforming raw data into meaningful business insights and decision-making support.



REFERENCES



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- Google Looker Studio: <https://lookerstudio.google.com/>
- Python Documentation: <https://docs.python.org/3/>
- Pandas Documentation: <https://pandas.pydata.org/docs/>
- Matplotlib Documentation: <https://matplotlib.org/stable/>
- Seaborn Documentation: <https://seaborn.pydata.org/>
- NumPy Documentation: <https://numpy.org/doc/>





THANK YOU

